

Lagzouli Riyad

Data Science Engineering Student — INSEA, Rabat

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PROFILE

Final-year Data Science engineering student at INSEA, focused on data engineering: ETL/ELT pipelines, lakehouse architectures (Databricks, Delta Lake, dbt), orchestration (Airflow, Databricks Jobs) and deterministic expert systems, with a solid background in statistical modeling and machine learning. Seeking a final-year internship (PFE) starting **February 1, 2027** to design and productionize reliable data pipelines.

TECHNICAL SKILLS

Machine Learning & Modeling: Random Forest, XGBoost, Regression, Classification, Clustering, SMOTE

Deep Learning & NLP: Transformers (BERT), Word Embeddings, Neural Networks, TF-IDF, Local LLM (Ollama)

Time Series & Statistics: Box-Jenkins Methodology, Stationarity Tests (ADF, KPSS), Statistical Inference

Decision Support & Expert Systems: Deterministic expert systems, rule-based inference engines, formal logic, model explainability

Data Engineering & Architecture: Databricks, PySpark, Delta Lake, dbt, Apache Airflow, DuckDB, PostgreSQL, PostGIS, Medallion Architecture

Databases & Query Optimization: Advanced SQL (Relational Modeling, Analytical Queries, Indexing), RDBMS

Development, Backend & DevOps: FastAPI, Docker, Git/GitHub, GitHub Actions (CI), Python, R

Business Intelligence & Reporting: Power BI, DAX Modeling, Interactive Executive Dashboard Design

PROFESSIONAL EXPERIENCE

Data Science & Engineering Intern June 15 – September 11, 2026

Bifolia — SmartFellah project, agronomic decision support

- Designed a deterministic expert system: every recommendation is traceable to the rules and thresholds that triggered it
- Built Airflow-orchestrated ETL pipelines: Ministry of Agriculture PDF documents structured into JSON (local LLM, Ollama), NDVI, NDWI and EVI satellite indices (Google Earth Engine) and OpenWeather forecasts, stored in PostgreSQL/PostGIS under Docker
- Developed a FastAPI REST API with 5 services: crop recommendation by municipality, daily irrigation and fertilization advice (FAO method) and 5-day disease alerts

Statistical Analysis Discovery Internship July 2025

High Commission for Planning (HCP) — Regional Directorate of Fez-Meknes

- Analyzed the National Survey on the Informal Sector to identify structural macroeconomic patterns and production unit behaviors
- Explored raw micro-datasets to ensure data reliability and consistency

PROJECTS

Morocco Bank Network Lakehouse 2026 – 2027

Databricks · PySpark · Delta Lake · dbt · GitHub Actions · Power BI View on GitHub

- Daily pipeline orchestrated with Databricks Jobs ingesting 2,700+ OpenStreetMap bank branches via API: Delta medallion architecture, idempotent incremental loading (MERGE) and SCD2 history
- Multilingual normalization (French, Arabic, Tifinagh) with 97.6% identification, spatial join to Morocco's 12 regions and branches-per-10,000-inhabitants KPI
- dbt star schema covered by 25 tests, GitHub Actions CI and Power BI dashboard

Bank Branch Customer Reviews Analysis 2025 – 2026

Python (Selenium, Pandas) · TextBlob · DuckDB · Power BI View on GitHub

- Google Maps review collection by web scraping, medallion layers, sentiment analysis and decision dashboard

Credit Card Fraud Detection 2025 – 2026

Python · Scikit-learn · Random Forest · XGBoost · SMOTE View on GitHub

- Machine learning pipeline comparing ensemble models (Random Forest, XGBoost) and handling extreme class imbalance with SMOTE

Time Series Analysis — Business Bankruptcies 2024 – 2025

R · Box-Jenkins (ARIMA/SARIMA) · Statistics View on GitHub

- Modeling quarterly business bankruptcies in Paris, stationarity testing (ADF, KPSS), and out-of-sample risk forecasting

NLP Spam Detection 2025 – 2026

Python · TF-IDF · BERT (Transformers) · scikit-learn View on GitHub

- Text classification comparing TF-IDF vs BERT embeddings for textual anomaly and fraud detection

EDUCATION

Engineering Degree in Data Science 2024 – Present

National Institute of Statistics and Applied Economics (INSEA), Rabat

University Diploma (DEUST) — Mathematics, CS, Physics 2022 – 2024

Faculty of Science and Technology, Fez (High School Diploma in Physical Sciences in 2022)

CERTIFICATIONS

IBM: Python for Data Science
NVIDIA: Fundamentals of Deep Learning
DataCamp: Supervised Learning (scikit-learn)
DataCamp: Unsupervised Learning in Python
DataCamp: Advanced Git

LEADERSHIP & ACTIVITIES

President of INSEA IT Club: LLM/Web workshops, negotiated 100 DataCamp licenses, NVIDIA workshop.

LANGUAGES

Arabic: Native
French: Fluent / Bilingual
English: Intermediate (B1)

INTERESTS

Travel · Sports · Music